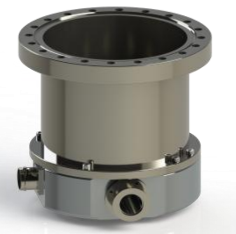
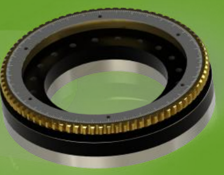
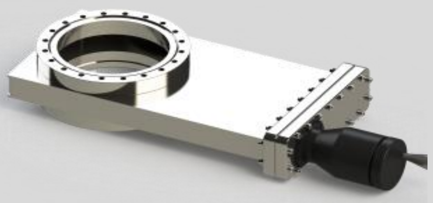
System Designs & Implementations

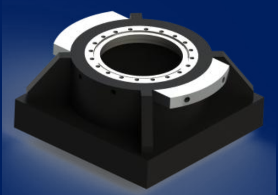




羅仁佑 Robert Jen-Iu Lo

- Windowless fluorescence absorption
- Photochemistry system in U9
- Absorption & Emission System In HF (03B)
- Gas Absorption System In HF (03B)
- Photodesorption, emission system
- Knudsen Cell
- Gas sample inlet
- VUV absorption system
- Microwave discharge lamp and absorption cell
- Stark effect system
- Atom gun system





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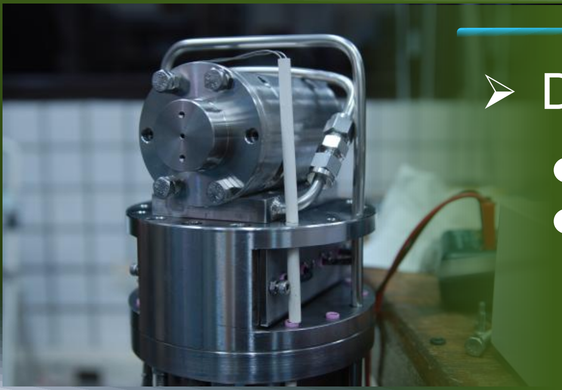
https://github.com/JenIuLo/JenIuLo-FP

Windowless fluorescence absorption system
--- temperature variation

Full System Assembly

- System Overview
  - LN cryo-head
  - Spectrometer
  - VUV (Differential pumping)
  - VUV flux detector
  - Solar blind PMT(160~320 nm)

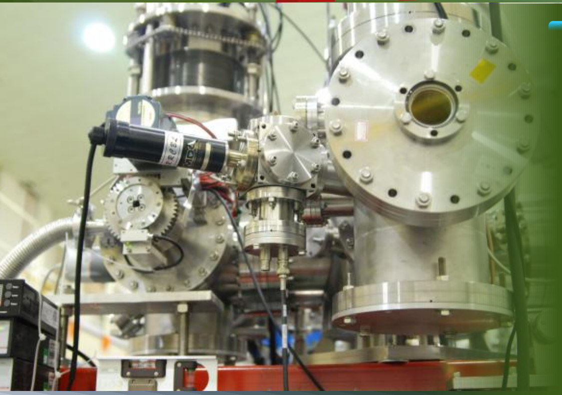


- Differential pumping system
  - 2 stages, 2 turbo
  - Lined 3 nozzle tubes
- Home made spectrometer
  - 5 stages MCPs
  - VUV grating
  - 1 turbo pump
- LN cryo-head
  - Cell (absorption & fluorescence)
  - Low temperature detector (diode 4 lines)
  - heater

Windowless fluorescence absorption system --- temperature variation

Full System Assembly with main frame

- System Overview
  - LN cryo-head
  - Spectrometer
  - VUV (Differential pumping)
  - VUV flux detector
  - Solar blind PMT(160~320 nm)



- Differential pumping system
  - 2 stages, 2 turbo
  - Lined 3 nozzle tubes
- Home made spectrometer
  - 5 stages MCPs
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- LN cryo-head
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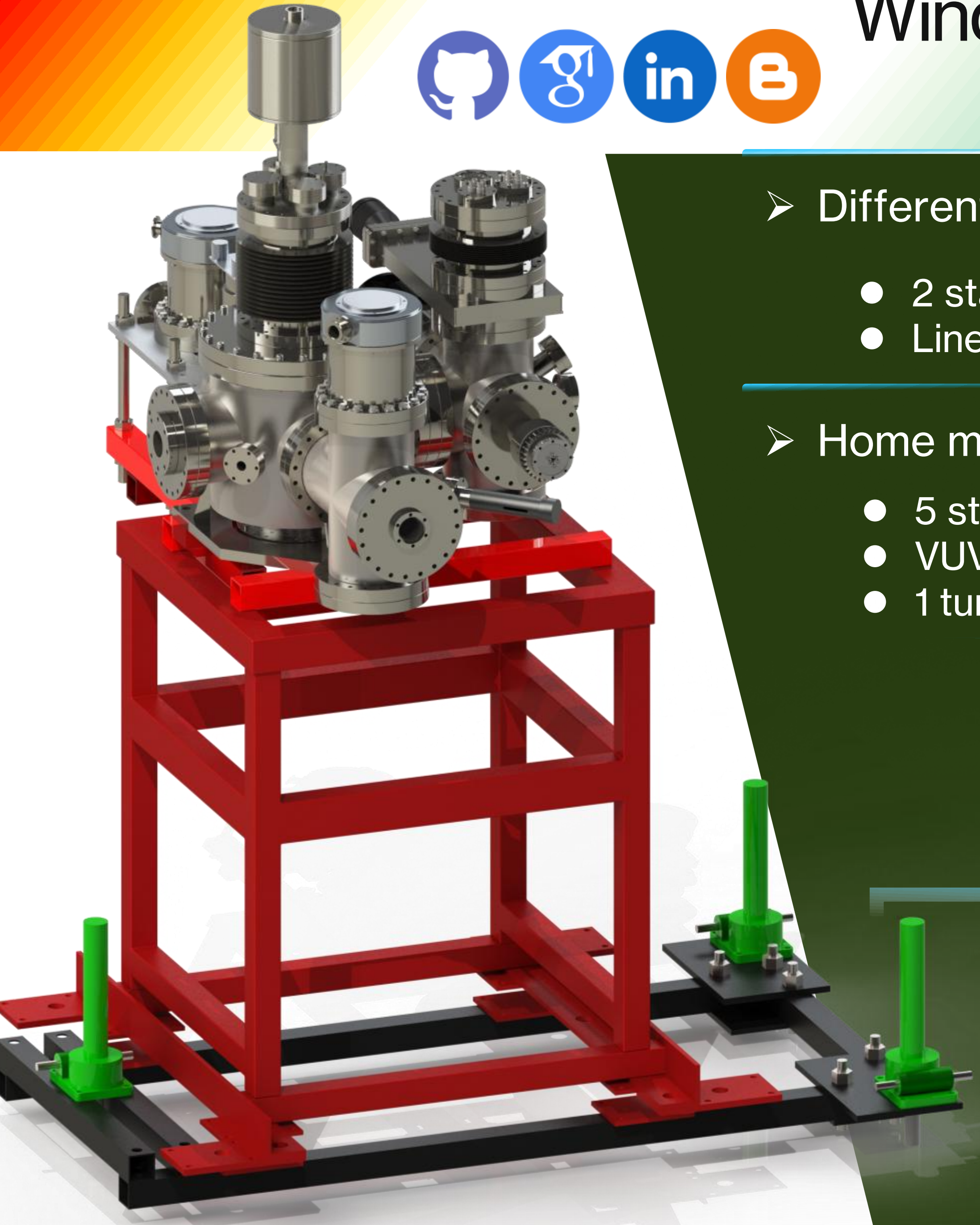
Windowless fluorescence absorption system
--- temperature variation



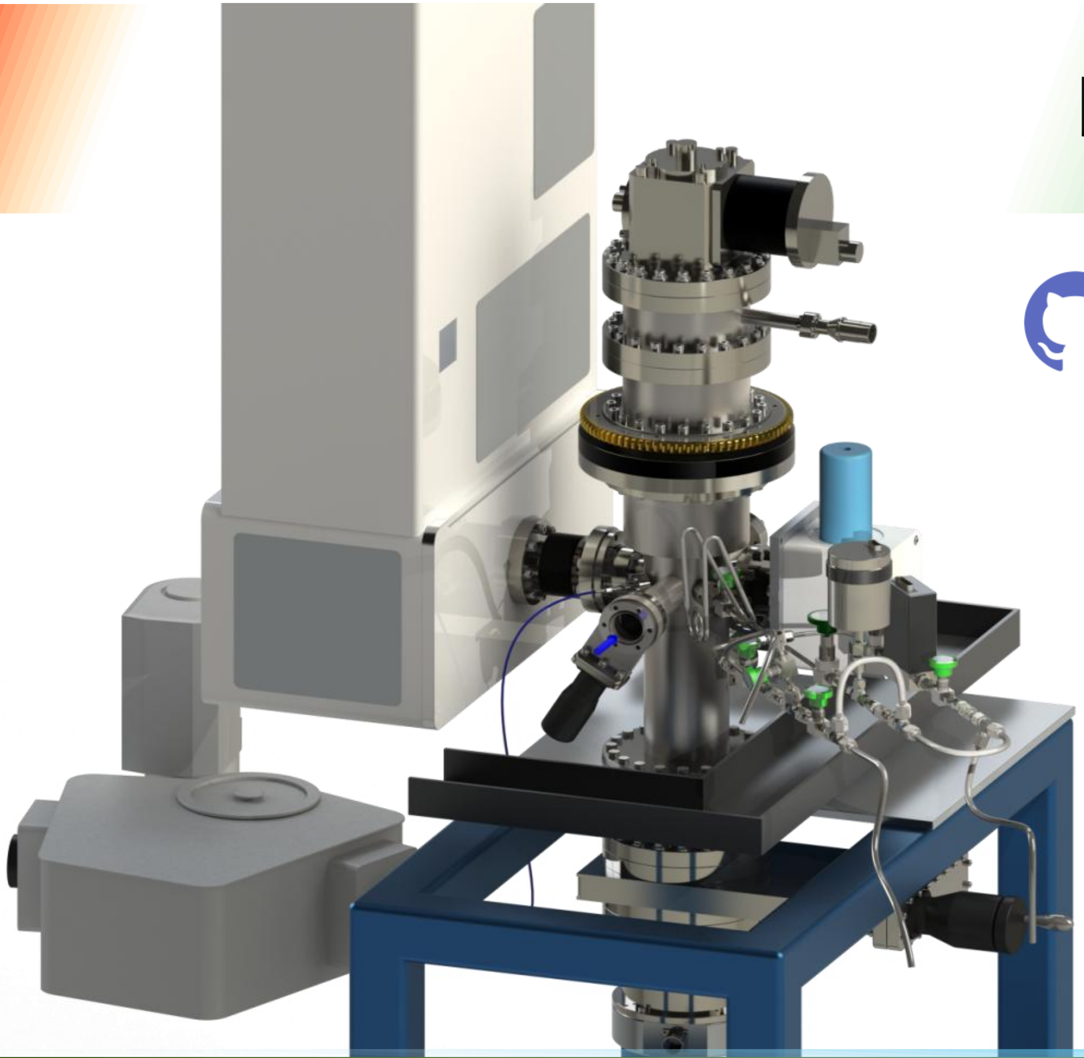
- Differential pumping system
  - 2 stages, 2 turbo
  - Lined 3 nozzle tubes
- Home made spectrometer
  - 5 stages MCPs
  - VUV grating
  - 1 turbo pump

Full System Assembly
extra frame for U9 (21B2)

- System Overview
  - LN cryo-head
  - Spectrometer
  - VUV (Differential pumping)
  - VUV flux detector
  - Solar blind PMT(160~320 nm)

- LN cryo-head
  - Cell (absorption & fluorescence)
  - Low temperature detector (diode 4 lines)
  - heater

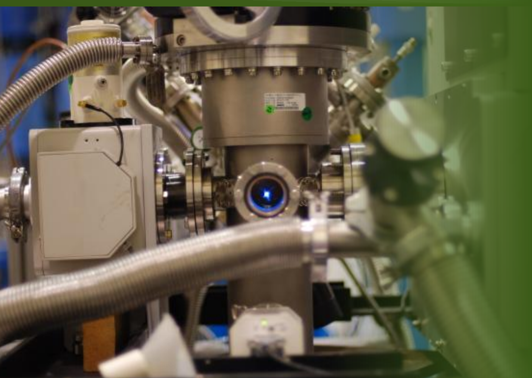
Photochemistry system in U9 (21A2)



➢ Front side view

➢ System Overview

- FTIR
- iHR320
- 4K cryo-head
- Gsa inlet
- RNN vacuum rotator



➢ 4K cryo-head

- Temp sensor (diode 4 lines)
- KBr substrate

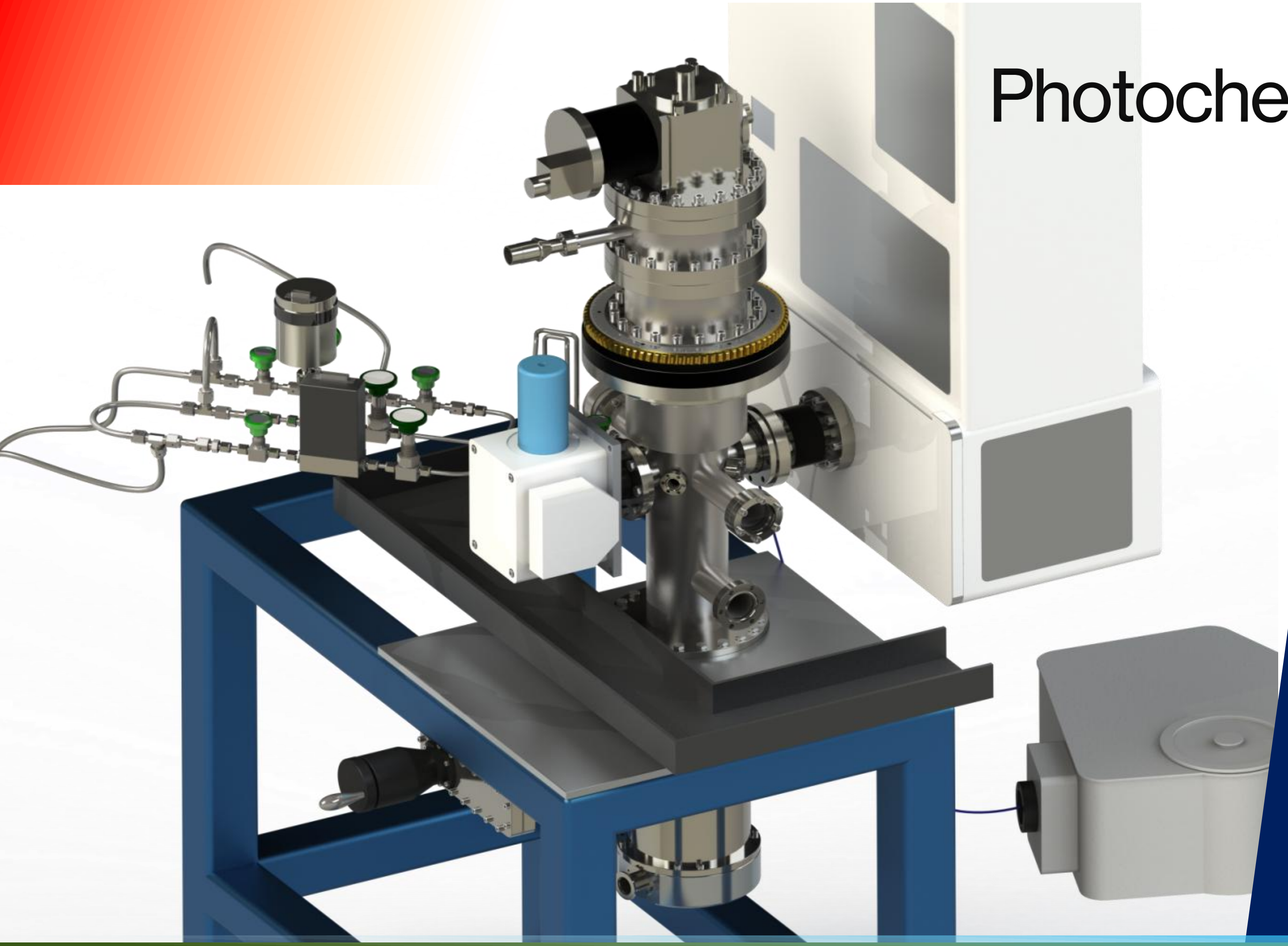
➢ FTIR

- Vacuum optical path
- IR detector

➢ iHR320

- 3 grating 600 1200 2400 l/mm
- CCD LN cooled
- Optical fiber+colleimator

Photochemistry system in U9 (21A2)



- End side view

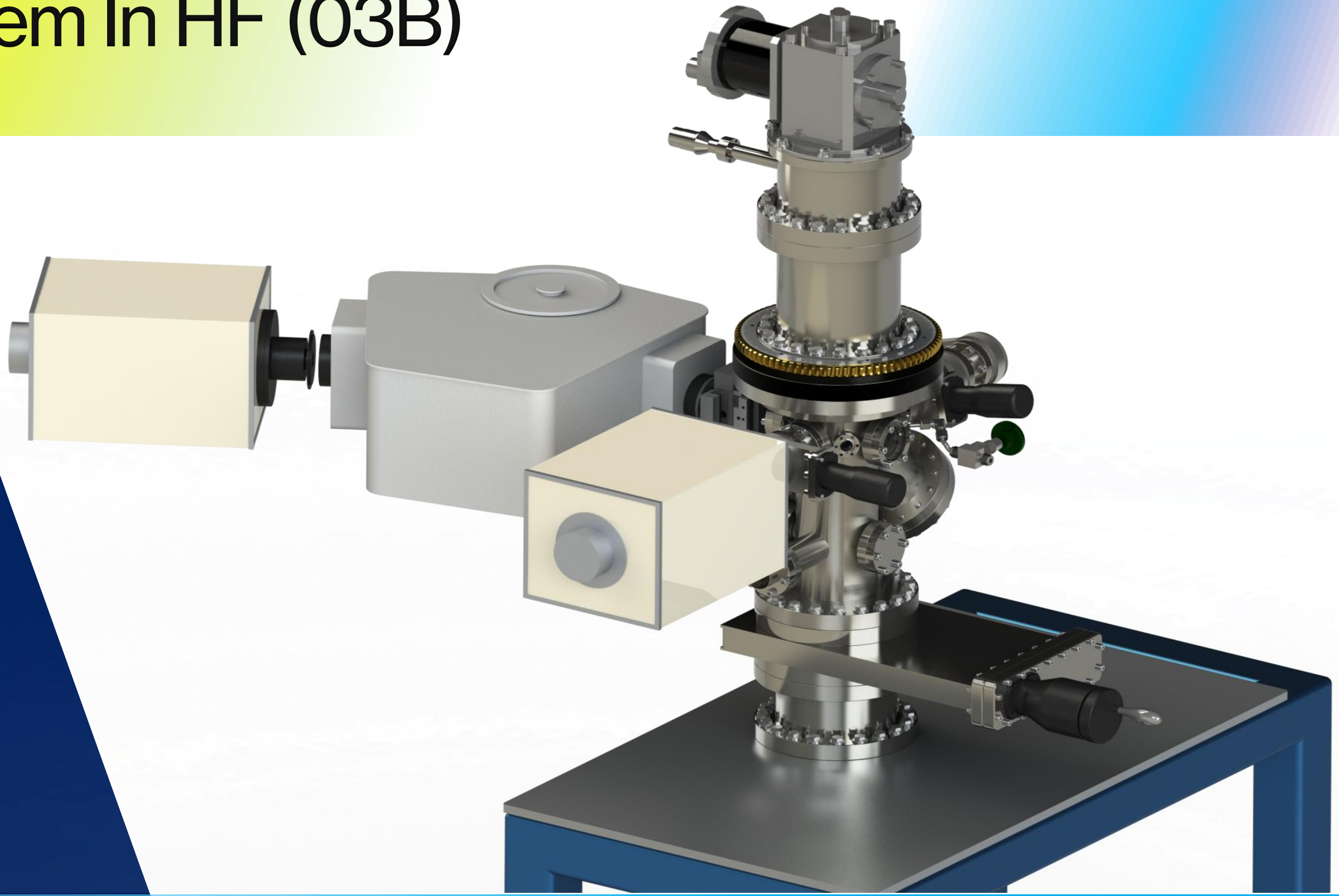
- System Overview
  - FTIR
  - iHR320
  - 4K cryo-head
  - Gsa inlet
  - RNN vacuum rotator

- 4K cryo-head
  - Temp sensor (diode 4 lines)
  - KBr substrate

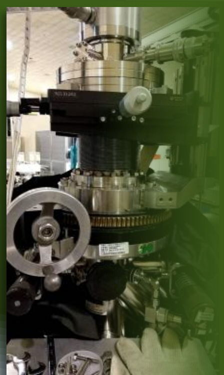
- FTIR
  - Vacuum optical path
  - IR detector

- iHR320
  - 3 grating 600 1200 2400 l/mm
  - CCD LN cooled
  - Optical fiber+colleimator

Absorption & Emission System In HF (03B)



- System Overview
  - BL03 HF
  - Solid state Exp.
  - iHR320
  - 4K cryo-head
  - Gsa inlet (deposed on LiF)
  - RNN vacuum rotator

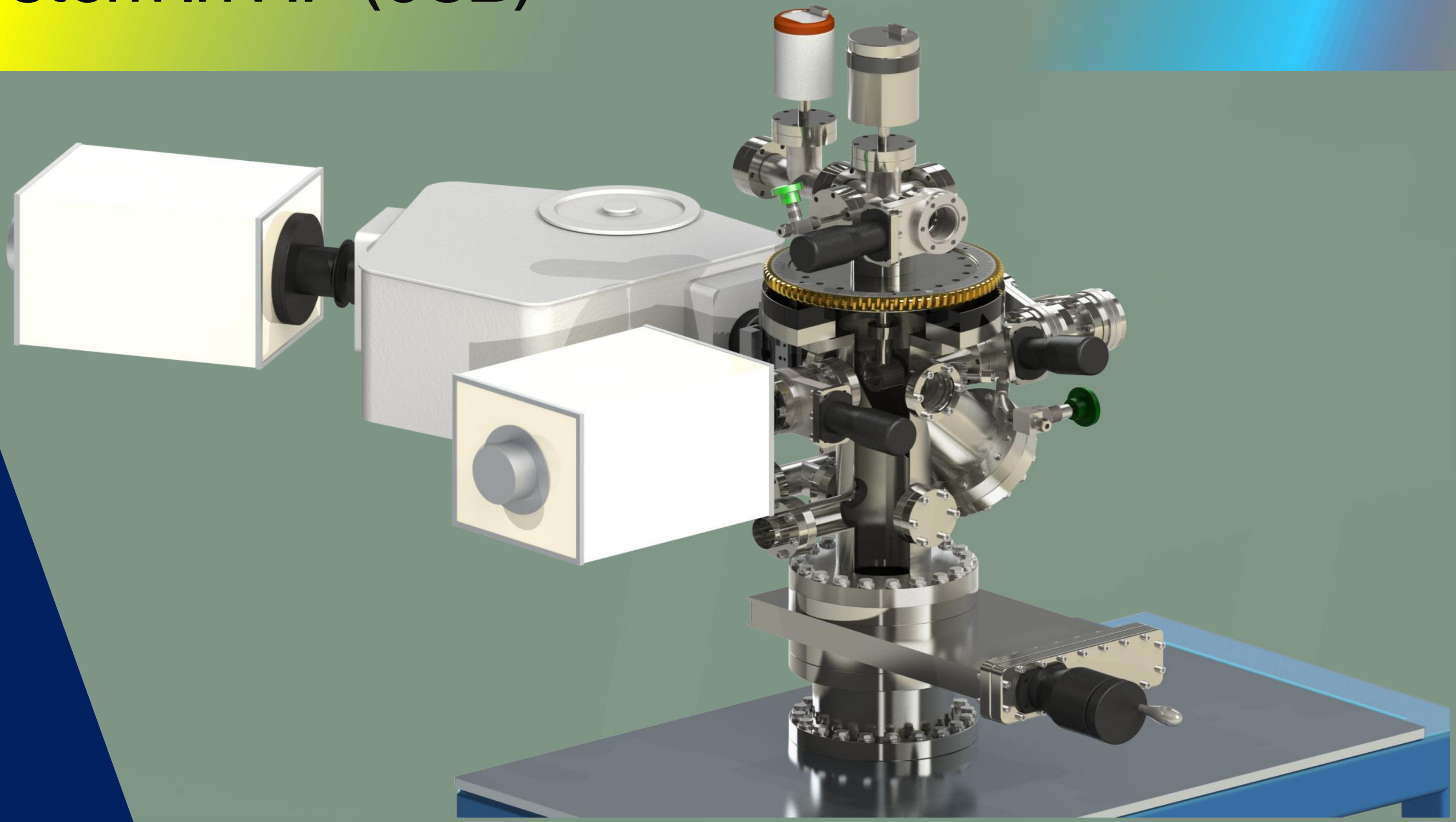


- 4K cryo-head
  - Temp sensor (diode 4 lines)
  - LiF substrate

- NSRRC
  - Hight Flux BL03
  - Branch B

- iHR320
  - 3 grating 600 1200 1800 l/mm
  - PMT TE cooled

Gas Absorption System In HF (03B)



- System Overview
  - BL03 HF
  - Absorption cell
  - iGsa inlet
  - RNN vacuum rotator
  - PMT TE cooling

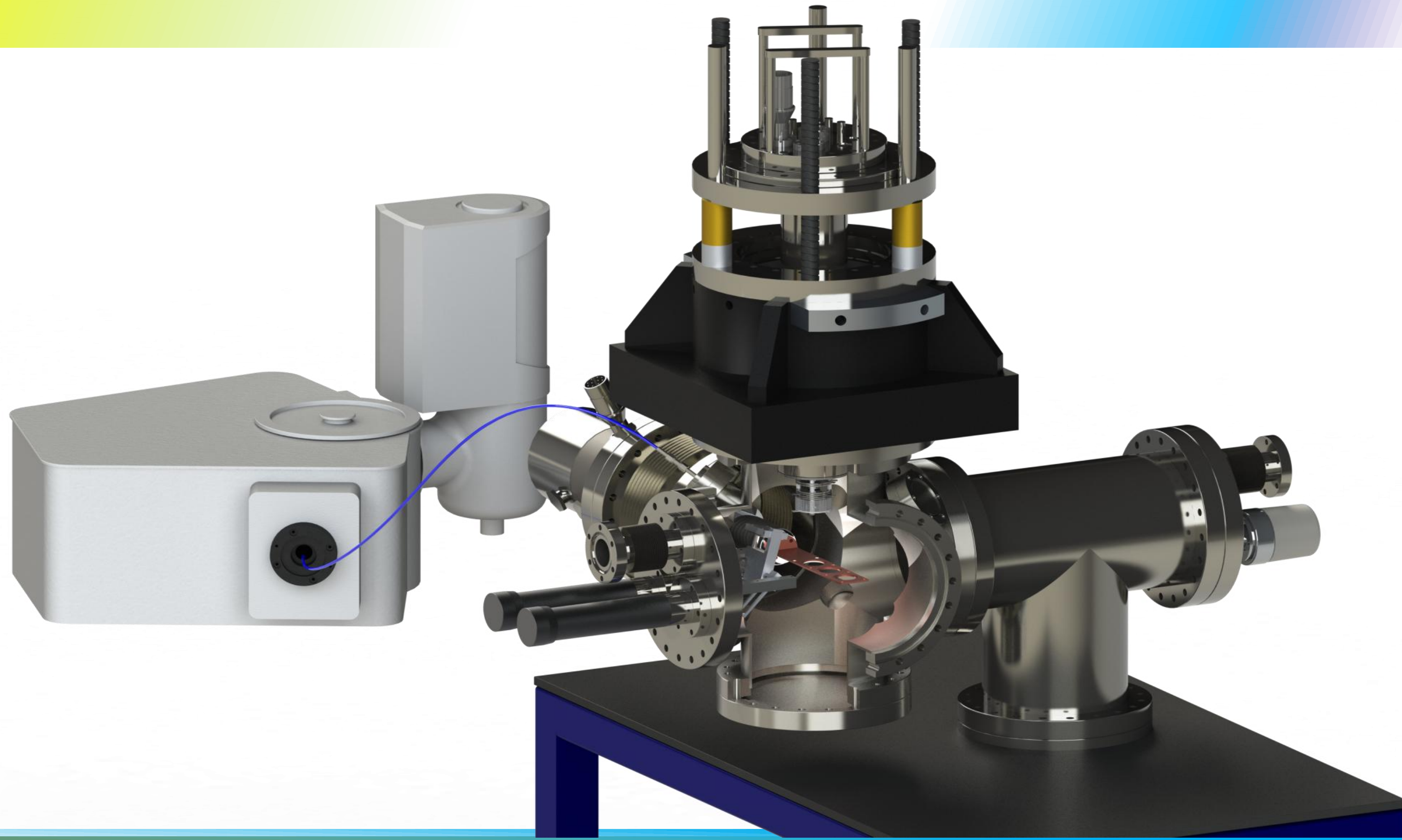
- Absorption cell
  - Length 10 cm
  - LiF vacuum windows

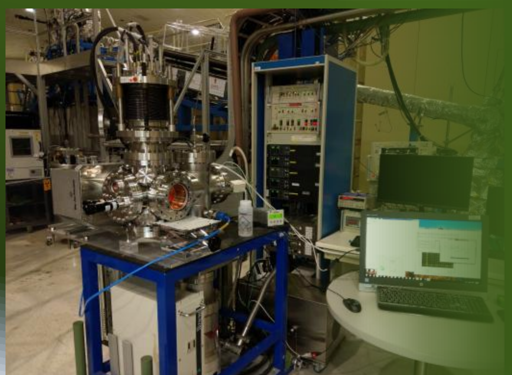
- NSRRC
  - Hight Flux BL03
  - Branch B

- iHR320
  - 3 grating 600 1200 1800 l/mm
  - PMT TE cooled

Photodesorption, emission system

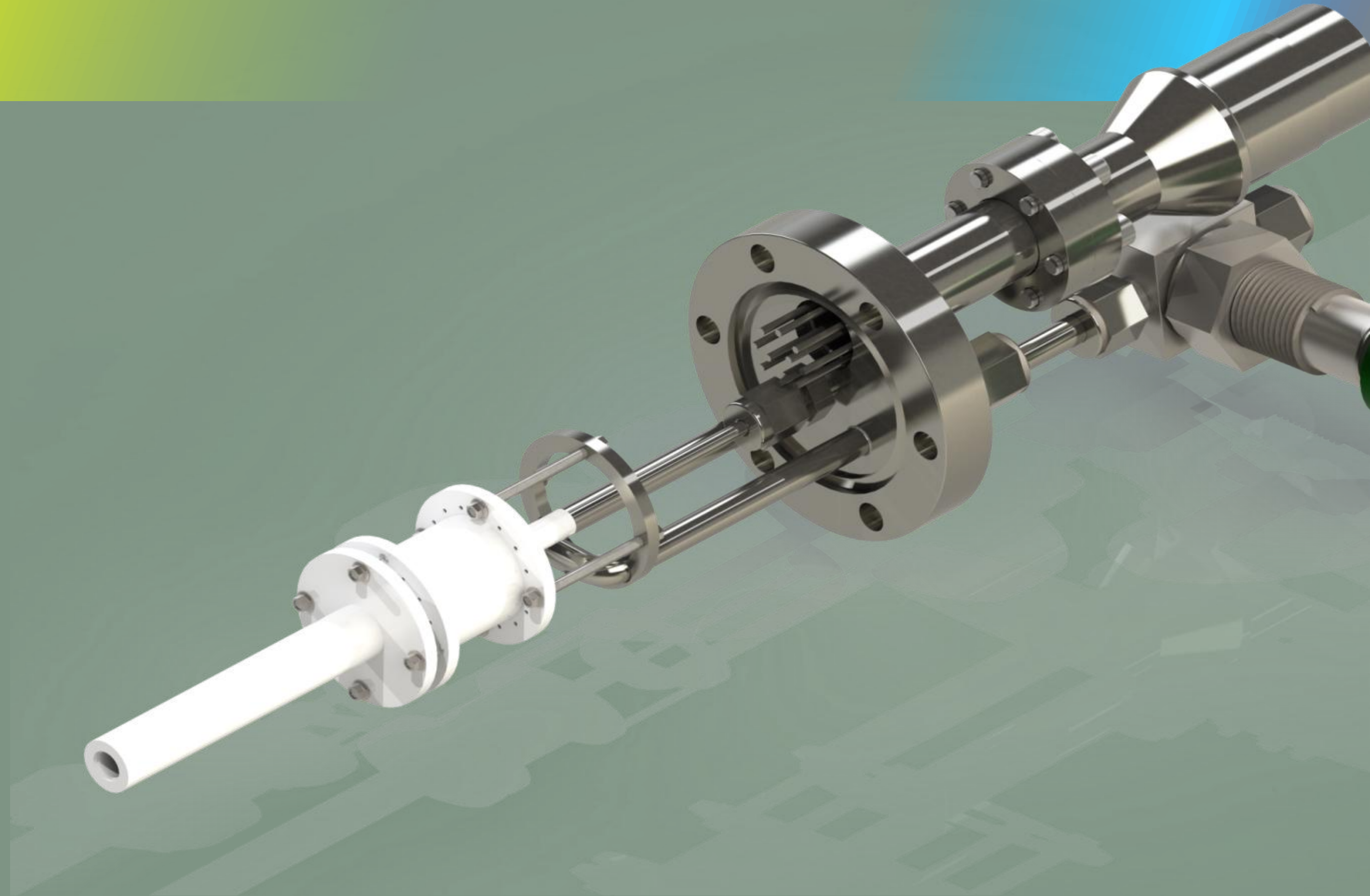
- System Overview
  - BL08B
  - iHR320
  - QMS
  - vacuum translator(Mass)
  - Golden mirror (reflecting VUV)





- Mass spectrum meter
  - QMS (quardruple)
  - 200 amu
  - Hight troughput
- NSRRC
  - BL08
  - Branch B
- iHR320
  - 3 grating 600 1200 2400 l/mm
  - CCD LN cooled

Knudsen Cell



- System Overview
  - Sample oven
  - Nozzle oven
  - Electric feedthrough
  - Buffer gas inlet
  - Adjustable valve

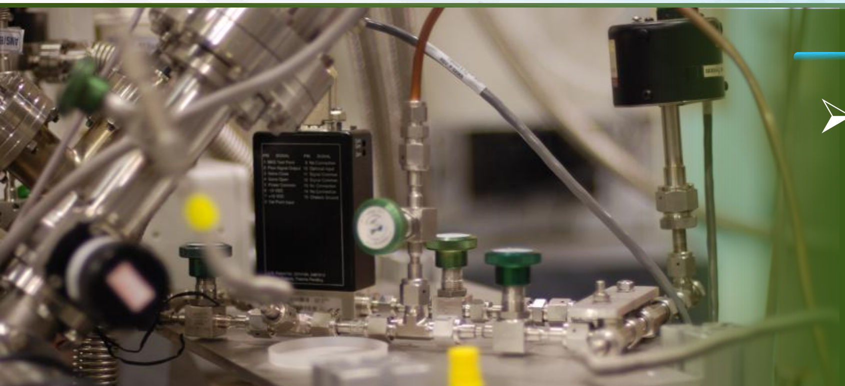
- Sample oven
  - 30 mm long
  - Diameter 20mm

- Nozzle oven
  - 10 cm long
  - Nozzle size 4mm

- Flage of holder
  - 2.75 in
  - 1.33 in feed through
  - 12 lines feed through

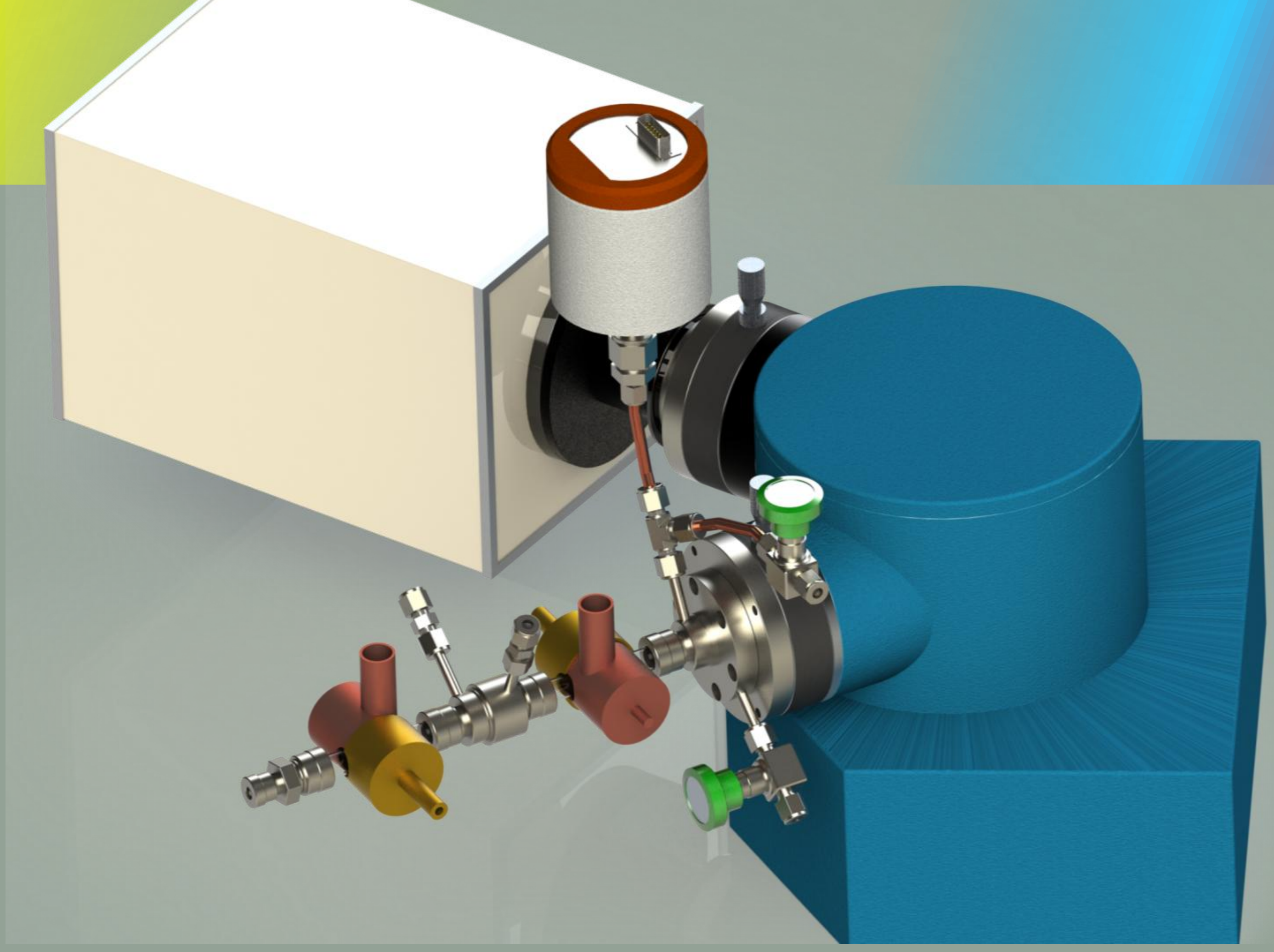
Gas sample inlet system for photochemistry

- System Overview
  - 4 on-off valves
  - 2 adjustable valves
  - Flow controller
  - Baratron pressure gauge



- swagelok
  - on-off valve
  - adjustable valve
- Flow controller
  - 10 sccm
  - Hand manual
- Baratron pressure gauge
  - 1000 Torr
  - 10 Torr

VUV absorption system



➢ System Overview
- VM502
- Absorption cell
- PMT
- F2 lamp
- Pressure gauge

➢ Discharge Lamp
- Micro-wave
- F2 gas

➢ PMT
- TE cooled
- Visible range
- sodium salicylate

➢ VM502
- VUV grating 1200 l/mm
- 200mm focusing length
- Vacuum monochromator

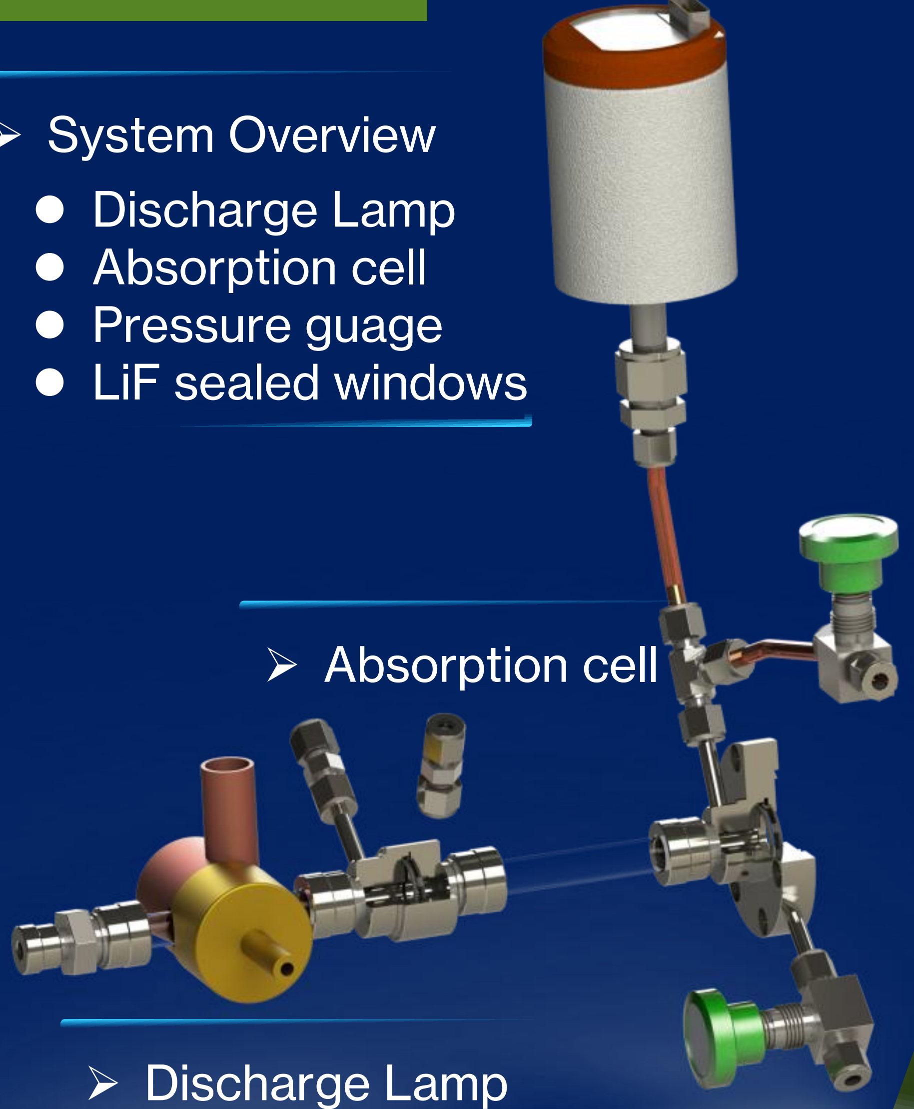
Microwave discharge lamp and absorption cell

➢ Perspective view

● Baratron pressure gauge

➢ System Overview

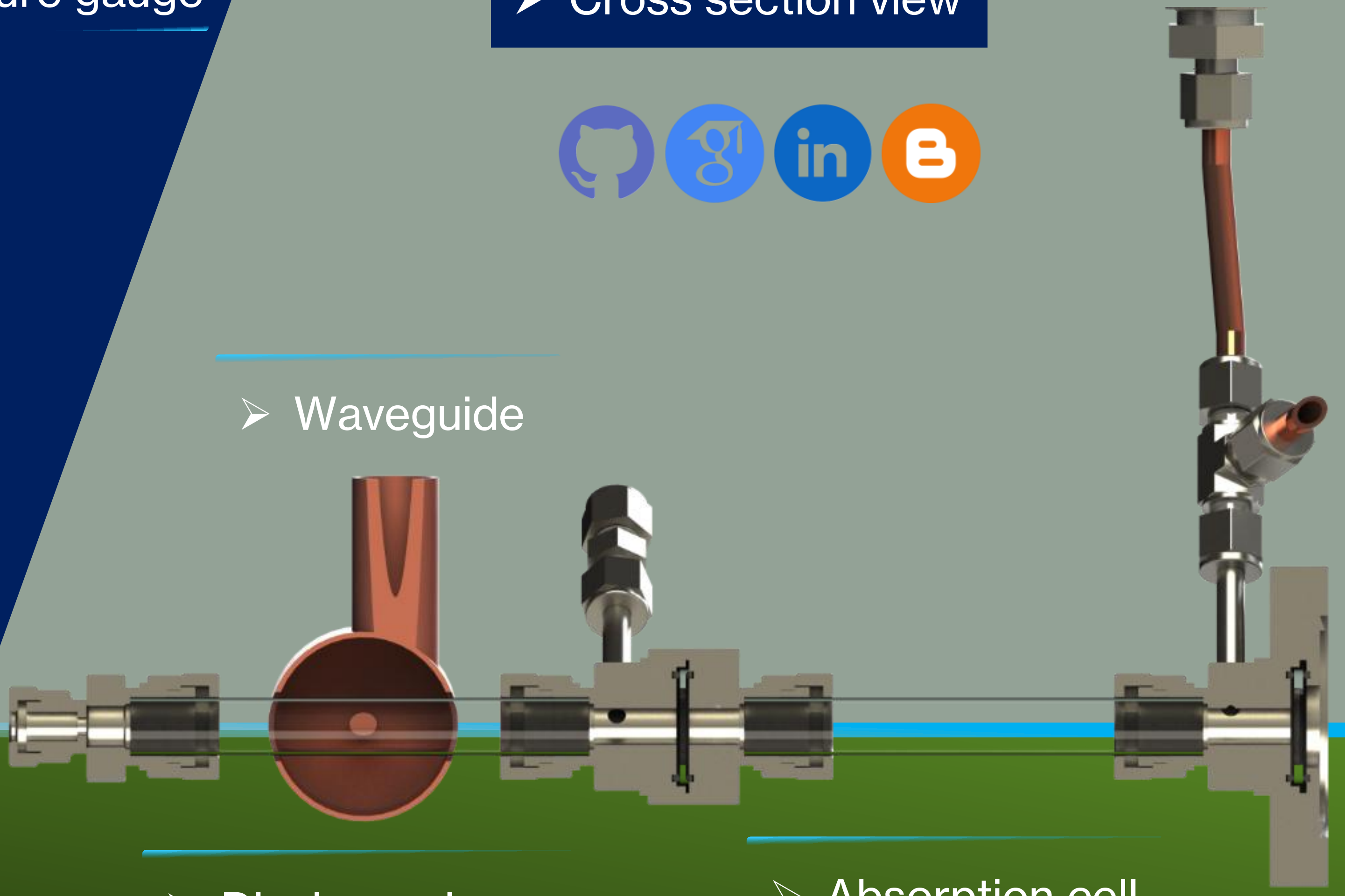
- Discharge Lamp
- Absorption cell
- Pressure guage
- LiF sealed windows



➢ Absorption cell

➢ Discharge Lamp

➢ Cross section view



➢ Waveguide

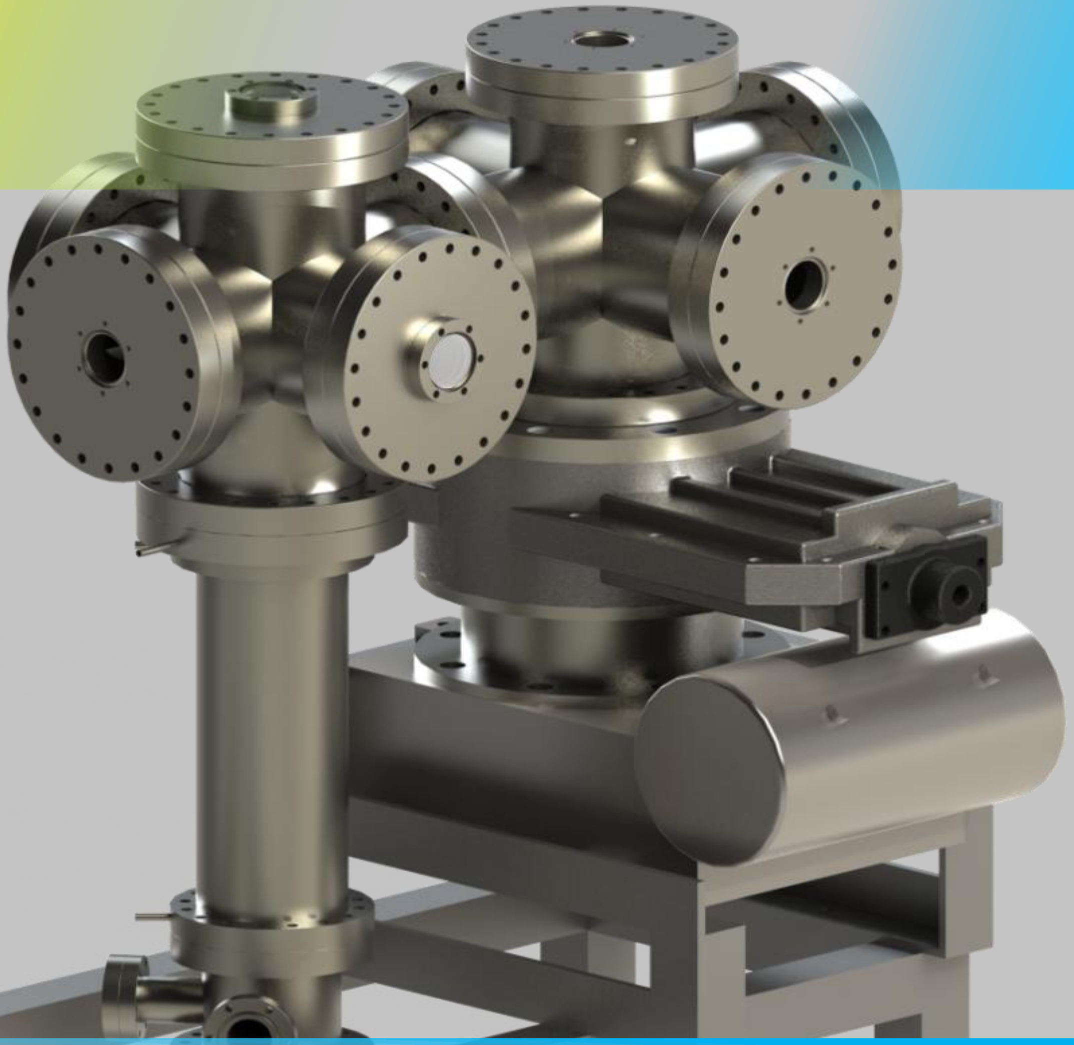
➢ Discharge Lamp

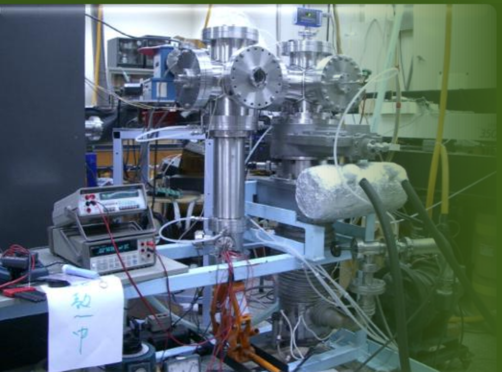
- Micro-wave
- F2

➢ Absorption cell

Stark effect system

- Experiences:
  - BL04 Seya
  - BL21B2
  - Li, Na
  - Mg, Ca, Sr, Ba

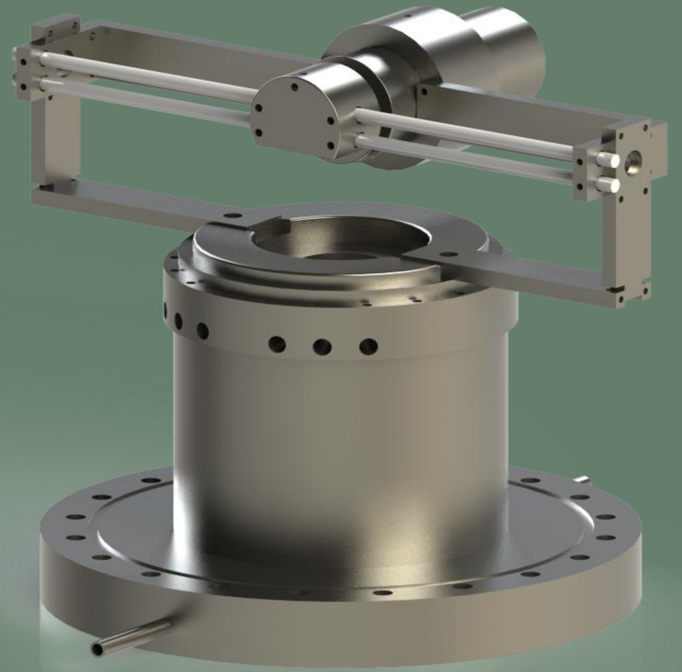




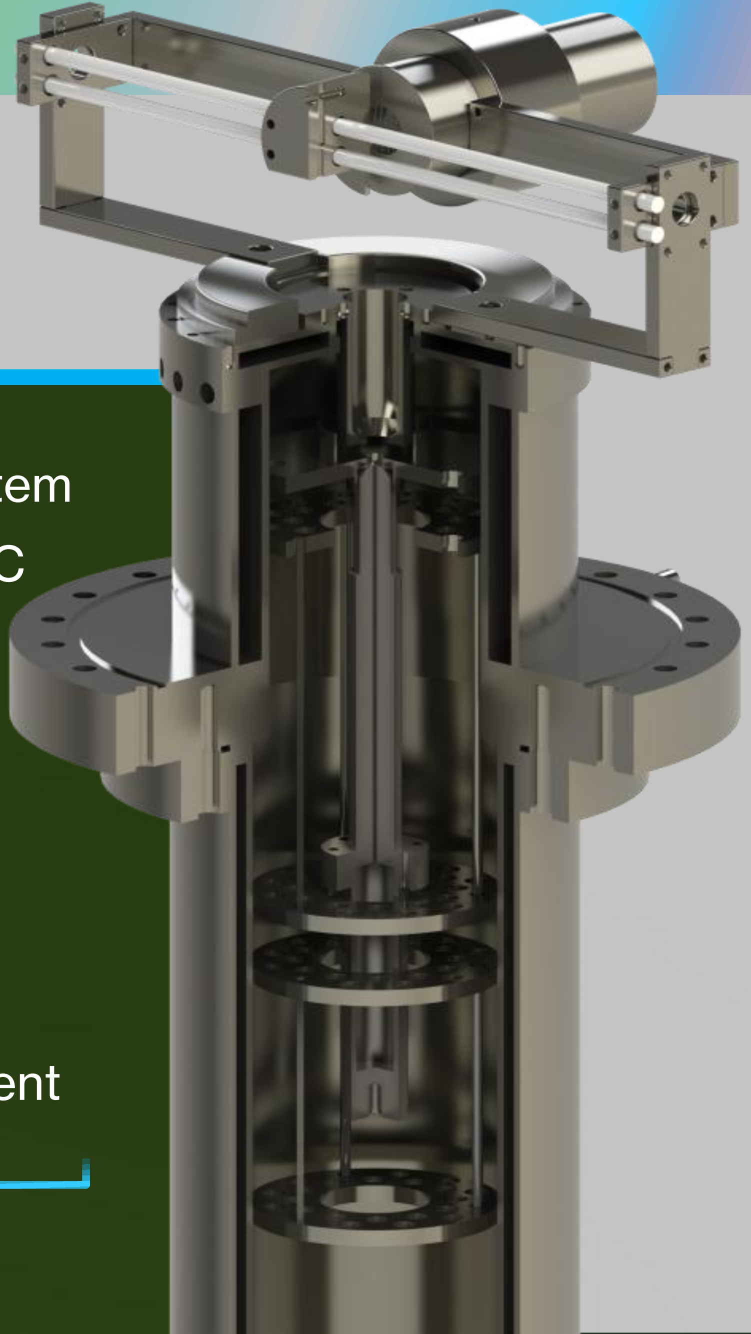
- Interaction zone
  - 5 mm gap
  - 25 KV
  - 50 KV/cm
- Ion detector
  - 2-stage MCP
  - Amp up to 10^5
- Atom gun
  - Temp 1200 C
  - Vacuum oven
  - Tunsten filerment
  - Water cooling

Atom gun system

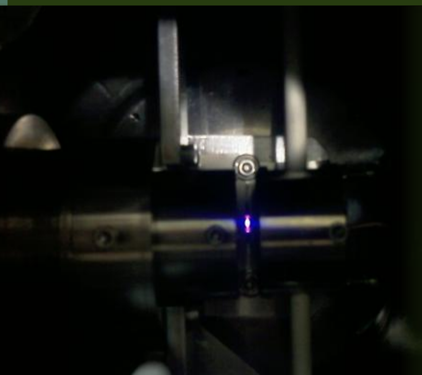
- perspective view



- Cross section view



- sample preparation system
  - Vacuume Oven 1200C
  - Radiation shielding
  - Cape Water cooling
  - Main water cooling
  - Atom beam shield
  - Atom gun
  - 200mm long nozzle
  - 1x2 mm CS of nozzle
  - Nozzle filement
  - Atom container filement
  - Interaction zone



- Interaction zone
  - 5 mm gap
  - 25 KV
  - On the top of atom gun system